

MINAMITORISHIMA, Japan

WMO: 479910

Lat: 24.290N

Lon: 153.979E

Elev: 22

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	64.5	65.8	48.3	50.1	66.5	50.2	53.9	67.3	27.6	70.0	25.7	69.9	13.1	10	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	89.3	79.6	88.5	79.4	87.8	79.1	81.4	86.4	80.8	86.0	80.3	85.5	10.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.0	155.9	84.4	79.3	152.3	84.3	78.8	149.5	84.2	45.1	86.5	44.4	86.0	43.9	85.7	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.3	24.5	22.3	DB	62.1	91.1	1.7	1.2	60.9	92.0	60.0	92.7	59.0	93.4	57.8	94.2	
			WB	53.9	82.2	1.5	1.0	52.8	83.0	51.9	83.5	51.1	84.1	50.0	84.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.8	72.3	71.5	73.0	76.3	79.4	82.8	83.5	83.3	83.5	82.6	80.0	76.4
	DBStd	5.05	2.78	3.04	3.23	3.01	2.55	1.94	1.85	1.88	1.73	1.52	2.05	2.80
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10496	693	602	713	790	912	984	1038	1032	1005	1010	899	818
	CDD65	5021	228	182	248	340	447	534	573	567	555	545	449	353
	CDH74	45634	468	346	707	1844	3739	6043	6873	6755	6626	6145	4076	2012
CDH80	12861	8	7	37	208	716	1988	2522	2424	2390	1814	620	127	
Wind	WSAvg	12.1	12.8	12.4	12.7	12.4	11.1	9.2	10.5	11.4	12.7	13.5	13.6	13.0
Precipitation	PrecAvg	41.1	2.8	1.7	2.5	2.1	2.9	2.0	6.1	6.2	3.7	4.3	2.8	3.7
	PrecMax	69.8	7.8	6.4	10.9	7.0	9.8	7.2	13.8	19.8	9.4	13.5	7.8	10.4
	PrecMin	19.4	0.6	0.3	0.1	0.0	0.1	0.1	0.5	1.2	1.0	0.4	0.5	0.4
	PrecStd	8.7	1.5	1.3	2.1	1.7	2.3	1.6	3.4	4.2	2.0	3.4	2.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.1	81.1	82.7	85.4	88.1	89.9	90.9	90.0	89.7	88.6	86.3	84.1
		MCWB	73.8	73.3	74.5	77.2	78.4	79.1	80.2	79.9	80.3	79.4	77.7	76.7
	2%	DB	79.4	79.3	81.0	83.8	86.5	88.7	89.4	88.8	88.4	87.3	85.2	82.7
		MCWB	72.3	72.1	73.5	76.1	77.9	78.9	79.6	79.7	79.8	78.9	77.1	75.5
	5%	DB	77.9	77.7	79.5	82.4	85.3	87.7	88.3	87.8	87.5	86.4	84.2	81.5
		MCWB	71.1	70.9	72.5	75.2	77.3	78.6	79.2	79.5	79.4	78.4	76.5	74.7
	10%	DB	76.5	76.1	77.8	81.0	83.9	86.7	87.2	86.9	86.7	85.5	83.1	80.2
		MCWB	70.0	69.7	71.4	74.3	76.6	78.3	78.8	79.2	79.1	78.0	76.0	73.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.7	75.2	75.8	78.2	80.0	81.5	81.8	82.3	81.9	81.4	81.3	78.1
		MCDB	79.3	78.8	80.8	83.7	85.5	86.9	88.1	86.7	86.9	85.9	83.4	82.1
	2%	WB	74.1	73.6	74.6	77.1	79.0	80.5	80.9	81.3	81.0	80.4	79.4	76.9
		MCDB	77.6	77.5	79.4	82.2	84.5	86.3	86.9	86.1	86.1	84.8	82.9	80.7
	5%	WB	72.8	72.4	73.6	76.2	78.2	79.8	80.2	80.6	80.5	79.7	78.1	76.1
		MCDB	76.7	76.4	78.1	80.8	83.4	85.5	86.1	85.7	85.6	84.4	82.4	80.0
	10%	WB	71.5	71.0	72.6	75.4	77.4	79.1	79.7	80.0	79.9	79.1	77.1	75.2
		MCDB	75.9	75.3	76.8	79.5	82.2	84.8	85.5	85.1	85.0	83.9	81.9	79.3
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.0	7.3	7.3	7.4	7.8	7.6	7.2	6.5	6.3	5.9	6.0
		MCDBR	7.6	8.7	9.0	8.9	9.0	8.5	8.7	7.9	7.2	6.9	6.7	6.8
		MCWBR	5.9	5.9	5.3	4.4	4.2	3.8	4.0	3.9	3.7	3.6	3.9	4.2
	5% WB	MCDBR	7.1	8.1	8.7	8.5	8.5	8.0	8.1	7.5	6.8	6.5	6.3	6.6
MCWBR		6.5	6.3	5.7	4.6	4.4	3.9	4.2	4.1	3.8	3.7	3.9	4.5	
Clear-Sky Solar Irradiance	taub	0.381	0.384	0.411	0.432	0.420	0.397	0.417	0.416	0.415	0.417	0.395	0.389	
	taud	2.520	2.492	2.423	2.416	2.487	2.567	2.517	2.522	2.519	2.490	2.537	2.524	
	Ebn at Noon	279	286	282	277	278	282	277	278	277	272	272	271	
	Edh at Noon	30	33	37	38	35	32	34	34	33	33	30	29	
All-Sky Solar Radiation	RadAvg	1181	1479	1738	1977	2095	2265	2025	1868	1861	1646	1365	1144	
	RadStd	80	89	98	148	165	106	132	215	120	63	66	64	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page